AWS SAA-C03

Practice Exam

Volume 1

Questions 1 to 70 — Foundations

IAM • EC2 • S3 • VPC • Databases • Serverless

Protodex Press

AWS SAA-C03 Practice Exam — Volume 1: Questions 1-70
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How to Use This Book

This is Volume 1 of a series covering all five SAA-C03 exam domains. It contains 70 scenario-based questions modeled on the format and difficulty you will encounter on the real exam.

How to study with it:
1. Attempt 10 questions in a single sitting. No notes. Time yourself: 1.5 minutes per question.
2. After 10 questions, check the answers section at the back. Read the explanation even for questions you got right — the reasoning is the value.
3. Flag questions where you guessed correctly. Revisit them in 48 hours.
4. Move to the next 10 questions only after you can explain WHY each answer in the previous batch is correct.

Domains covered in this volume:
• IAM and Security (Q1-Q12)
• EC2 and Compute (Q13-Q24)
• S3 and Storage (Q25-Q36)
• VPC and Networking (Q37-Q48)
• Databases (Q49-Q60)
• Application Integration & Serverless (Q61-Q70)

Each question has exactly one BEST answer. On the real exam, two options will often be technically correct — pick the one that best matches AWS's recommended pattern.

Good luck. The exam rewards careful reading more than memorization.

Volume 1 Question Breakdown

Total questions in this volume: 70

By domain:

- IAM: 12 questions (17%)
- EC2: 12 questions (17%)
- S3: 12 questions (17%)
- VPC: 12 questions (17%)
- DB: 12 questions (17%)
- Serverless: 10 questions (14%)

SAA-C03 official domain weights (for reference):

- Design Secure Architectures: 30%
- Design Resilient Architectures: 26%
- Design High-Performing Architectures: 24%
- Design Cost-Optimized Architectures: 20%

Vol. 2 (Q71-140) goes deeper into security and resilience. Vol. 3 (Q141-210) covers cost optimization. Vols. 4-5 finish the 350-question target.

Study Weighting Guide

The AWS Certified Solutions Architect - Associate (SAA-C03) exam tests your knowledge across four official domains, each carrying a specific percentage weight. To study effectively using the foundations in this volume, you should align your preparation time with these official proportions.

Here is how the 70 questions in this book map to the official exam domains, along with study recommendations:

Domain 1: Design Secure Architectures (30% of exam)

• Relevant Questions: Q1-Q12 (IAM & Identity), Q25-Q28 & Q33 (S3 Security & KMS).
• Focus Area: Ensure you understand cross-account access, IAM policies, envelope encryption with KMS, and security group vs. NACL differences.

Domain 2: Design Resilient Architectures (26% of exam)

• Relevant Questions: Q37-Q48 (VPC Networking & NAT), Q49-Q54 (RDS & Aurora Multi-AZ).
• Focus Area: Study high-availability networking patterns, NAT gateways across multiple AZs, and disaster recovery strategies.

Domain 3: Design High-Performing Architectures (24% of exam)

• Relevant Questions: Q13-Q24 (EC2 Compute Options), Q55-Q60 (DynamoDB & ElastiCache), Q61-Q70 (Lambda & Serverless scaling).
• Focus Area: Match compute types to performance requirements, understand database read/write scaling, and review Lambda cold starts.

Domain 4: Design Cost-Optimized Architectures (20% of exam)

• Relevant Questions: Q29-Q32 & Q34-Q36 (S3 lifecycle policies and transition rules), EC2 Spot vs. Reserved pricing.
• Focus Area: Automate object lifecycle transfers to colder storage tiers and size EC2 instances efficiently.

Question 1 IAM

A company runs a fleet of EC2 instances that must read objects from a private S3 bucket. The security team forbids storing long-lived credentials on the instances. Which approach meets the requirement with the LEAST operational overhead?

A. Create an IAM user, generate access keys, and store them in /etc/aws/credentials on each instance.

B. Create an IAM role with an S3 read policy and attach it to the EC2 instance profile.

C. Use AWS Secrets Manager to push credentials to each instance every 60 minutes.

D. Federate the EC2 instances through AWS IAM Identity Center using SAML 2.0.

Question 2 IAM

A solutions architect needs to grant a third-party auditor temporary read-only access to AWS resources for 7 days. Which approach is MOST secure?

A. Create an IAM user with ReadOnlyAccess and delete it after 7 days.

B. Share the root account credentials and rotate the password after the audit.

C. Create an IAM role with ReadOnlyAccess and a trust policy allowing the auditor's AWS account to assume it; set a 1-hour session duration.

D. Create access keys with ReadOnlyAccess and email them to the auditor.

Question 3 IAM

Developers complain that they cannot launch EC2 instances even though they have ec2:RunInstances permission. The CloudTrail event shows AccessDenied on iam:PassRole. What is the cause?

A. ec2:RunInstances also requires kms:Decrypt to launch.

B. The user lacks iam:PassRole permission for the instance profile being attached.

C. The default VPC was deleted.

D. The instance profile is in a different region.

Question 4 IAM

Which condition key allows an S3 bucket policy to restrict access to requests coming through a specific VPC endpoint?

A. aws:SourceIp

B. aws:SourceVpc

C. aws:SourceVpce

D. aws:PrincipalOrgID

Question 5 IAM

A company has 200 developers across 12 AWS accounts under AWS Organizations. They want a single sign-on solution with permission sets that map to job functions. Which service should they use?

A. AWS Cognito User Pools

B. AWS IAM Identity Center (formerly SSO)

C. Amazon Directory Service for Microsoft AD only

D. AWS Resource Access Manager

Question 6 IAM

An application running on Amazon ECS Fargate must call DynamoDB. Which IAM role provides the credentials to the application code?

A. Task execution role

B. Task role

C. Service-linked role

D. EC2 instance role

Question 7 IAM

A bucket policy uses 'Effect: Deny' with condition 'aws:SecureTransport: false'. What does this achieve?

A. Blocks all bucket access entirely.

B. Denies access when the request is made over HTTP instead of HTTPS.

C. Denies all access from outside the VPC.

D. Forces objects to be encrypted at rest.

Question 8 IAM

Which is NOT a valid component of an IAM policy?

A. Effect
B. Principal
C. TargetGroup
D. Condition

Question 9 IAM

A user has explicit Allow for s3:GetObject in an inline policy. The bucket policy has an explicit Deny for s3:GetObject on the same prefix. What happens when the user calls GetObject?

A. Allowed — inline IAM policy wins.

B. Denied — explicit Deny always wins.

C. Allowed for the first request, then denied.

D. Depends on the evaluation order specified in the policy.

Question 10 IAM

A new EC2 instance must access a secret stored in AWS Secrets Manager. The secret is encrypted with a customer-managed KMS key. Which permissions does the instance role need?

A. secretsmanager:GetSecretValue only

B. secretsmanager:GetSecretValue and kms:Decrypt on the KMS key

C. secretsmanager:GetSecretValue and kms:GenerateDataKey

D. Only IAM:PassRole

Question 11 IAM

Which IAM feature reports which permissions a role has actually used vs which are granted?

A. IAM Credential Report

B. IAM Access Advisor

C. AWS Trusted Advisor

D. IAM Policy Simulator

Question 12 IAM

A company wants to allow only EC2 instances tagged Environment=Prod to be terminated by a specific IAM group. Which approach works?

A. Add a Resource ARN filter — not possible with tags.

B. Use a Condition block with aws:ResourceTag/Environment: Prod on ec2:TerminateInstances.

C. Set a tag-based SCP at the account level.

D. Use IAM Access Analyzer rules.

Question 13 EC2

An e-commerce site has predictable steady traffic during business hours and unpredictable spikes during marketing campaigns. Which pricing combination MINIMIZES cost?

A. 100% On-Demand Instances

B. 100% Reserved Instances for peak capacity

C. Reserved/Savings Plans for baseline + On-Demand or Spot for spikes

D. 100% Spot Instances

Question 14 EC2

Which placement group strategy is BEST for an HPC workload requiring low-latency, high-throughput inter-node communication?

A. Cluster placement group

B. Spread placement group

C. Partition placement group

D. No placement group

Question 15 EC2

An EC2 instance reports 'Insufficient capacity' when starting from Stopped state. What is the MOST likely cause?

A. The AMI was deleted.

B. AWS does not have free capacity for that instance type in that AZ at that moment.

C. The IAM role expired.

D. The EBS volume is in a different region.

Question 16 EC2

Which EBS volume type is BEST for a database workload requiring 32,000 IOPS sustained with single-digit-millisecond latency?

A. gp2

B. gp3

C. io2 Block Express

D. st1

Question 17 EC2

What happens to data on an instance store volume when an EC2 instance is stopped?

A. Persists; reattaches on start.

B. Snapshotted to S3 automatically.

C. Lost permanently.

D. Encrypted and retained for 24 hours.

Question 18 EC2

An auto-scaling group must respond to traffic within 30 seconds. CloudWatch alarms take ~2 minutes. Which scaling policy is BEST?

A. Step scaling

B. Target tracking

C. Predictive scaling

D. Scheduled scaling

Question 19 EC2

Which feature provides a hardware-isolated EC2 environment for sensitive workloads, with no shared CPU sockets?

A. Dedicated Instance

B. Dedicated Host

C. Bare Metal Instance

D. Nitro Enclaves

Question 20 EC2

An application needs to share a file system with concurrent read/write access from 1000 EC2 instances across 3 AZs. Which storage option is MOST appropriate?

A. EBS Multi-Attach

B. Amazon EFS Standard

C. Amazon FSx for Windows

D. S3 with file gateway

Question 21 EC2

Spot Instance Advisor reports a frequency-of-interruption of '>20%' for m5.large in your region. What does this imply?

A. The instance type is unavailable.

B. Roughly 1-in-5 instances will be reclaimed within a typical Spot lifespan; suitable only for fault-tolerant workloads.

C. The on-demand price is 20% higher than typical.

D. Capacity is reserved for 20% of customers.

Question 22 EC2

Which AMI deployment pattern enables 'golden image' immutable infrastructure?

A. Launch base AMI, run config management on boot.

B. Build pre-baked AMI with all software via EC2 Image Builder, version it, deploy unchanged.

C. SSH into running instance and update packages weekly.

D. Run AWS Systems Manager Patch Manager daily.

Question 23 EC2

An Auto Scaling group spans 3 AZs with min=3, max=12, desired=6. One AZ fails. What does AWS do?

A. Marks desired as 4 to match surviving AZs.

B. Launches replacement capacity in the surviving AZs to maintain desired=6.

C. Waits for the failed AZ to recover.

D. Terminates all instances in the failed AZ permanently.

Question 24 EC2

What is the recommended way to update the OS on a production EC2 fleet without downtime?

A. SSH into each instance and run yum update during off-hours.

B. Bake a new AMI, update the launch template, and use ASG instance refresh.

C. Stop and start each instance to apply patches.

D. Restart the underlying hypervisor.

Question 25 S3

A media company stores 50 TB of video files. Files <30 days old are accessed daily; older files are accessed quarterly; files >180 days old are rarely accessed but must be retrievable within minutes. Which lifecycle policy minimizes cost?

A. S3 Standard → S3 Glacier Deep Archive at 30 days

B. S3 Standard → S3 Standard-IA at 30 days → S3 Glacier Flexible Retrieval at 180 days

C. S3 Standard → S3 Intelligent-Tiering immediately

D. S3 One Zone-IA forever

Question 26 S3

Which S3 feature prevents object deletion or overwriting for a specified retention period?

A. Versioning

B. MFA Delete

C. S3 Object Lock

D. Cross-Region Replication

Question 27 S3

An application uploads 50 GB files to S3 from on-premises and frequently fails partway through. Which approach improves reliability?

A. Increase bucket bandwidth quota.

B. Use S3 multipart upload to split into parts; retry only failed parts.

C. Use S3 Select to filter before upload.

D. Switch to S3 One Zone-IA for upload.

Question 28 S3

S3 Standard offers what durability and availability?

A. 99.9% durability, 99.99% availability

B. 99.99% durability, 99.95% availability

C. 99.999999999% (11 nines) durability, 99.99% availability

D. 99.999999999% durability, 99.9% availability

Question 29 S3

A bucket holds objects with sensitive PII. Which combination provides defense-in-depth?

A. Public read access + SSE-S3

B. Block Public Access + SSE-KMS with CMK + bucket policy enforcing aws:SecureTransport + CloudTrail data events

C. Bucket policy allowing all VPCs + SSE-S3

D. Object Lock alone

Question 30 S3

Which storage class has a 30-day minimum storage charge?

A. S3 Standard

B. S3 Standard-IA

C. S3 One Zone-IA

D. Both B and C

Question 31 S3

What is the MOST appropriate use case for S3 Transfer Acceleration?

A. Uploading from EC2 in the same region as the bucket.

B. Uploading large files from distant geographies via CloudFront edge locations.

C. Replicating across regions.

D. Speeding up downloads from S3 to mobile apps.

Question 32 S3

Which is required to enable Cross-Region Replication (CRR) on an S3 bucket?

A. Versioning on source AND destination buckets

B. Object Lock enabled

C. Same AWS account for both buckets

D. Identical bucket names in both regions

Question 33 S3

Which encryption option requires AWS to call your KMS endpoint on every GET, possibly affecting throughput at scale?

A. SSE-S3

B. SSE-KMS

C. SSE-C

D. Client-side encryption

Question 34 S3

How can you query CSV objects stored in S3 without loading them into a database or EC2?

A. S3 Object Lambda

B. S3 Select / Athena

C. S3 Inventory

D. S3 Storage Lens

Question 35 S3

A request to S3 returns HTTP 503 SlowDown. What is the recommended response?

A. Switch to S3 One Zone-IA

B. Implement exponential backoff retry; consider distributing keys across prefixes for higher TPS.

C. Upgrade the AWS support plan

D. Lower the request rate to <100/sec permanently

Question 36 S3

Which feature provides a single endpoint that intelligently routes to the lowest-latency or healthiest S3 bucket across regions?

A. Cross-Region Replication

B. S3 Multi-Region Access Points

C. CloudFront with S3 origin

D. S3 Transfer Acceleration

Question 37 VPC

Which VPC component allows EC2 instances in a PRIVATE subnet to reach the public internet for OS updates while preventing inbound connections from the internet?

A. Internet Gateway
B. NAT Gateway
C. VPC Peering
D. Transit Gateway

Question 38 VPC

What is the MAIN difference between a Security Group and a Network ACL?

A. Security Groups are at the subnet level; NACLs are at the instance level.

B. Security Groups are stateful and allow-only; NACLs are stateless and support deny rules.

C. Security Groups are free; NACLs cost extra.

D. NACLs can reference other security groups by ID.

Question 39 VPC

Two VPCs in the same region must communicate privately. The CIDR blocks overlap. Which solution works?

A. VPC Peering

B. Transit Gateway

C. VPC Peering with overlapping CIDR is impossible; re-IP one VPC or use PrivateLink for specific services.

D. AWS Direct Connect

Question 40 VPC

Which type of VPC endpoint is used for S3 and DynamoDB?

A. Interface endpoint
B. Gateway endpoint
C. Gateway Load Balancer endpoint
D. Transit Gateway endpoint

Question 41 VPC

A NAT Gateway is deployed in a single AZ. The application spans 3 AZs in private subnets. What is the risk?

A. Outbound traffic crosses AZ boundaries (data transfer charges) and the NAT becomes a single point of failure.

B. Security groups stop working.

C. Reserved Instance pricing is invalidated.

D. Route 53 fails to resolve.

Question 42 VPC

Which is the FASTEST connectivity option between on-premises and AWS for sustained 10 Gbps traffic?

A. Site-to-Site VPN

B. AWS Direct Connect

C. VPC Peering

D. Transit Gateway

Question 43 VPC

An application sends UDP traffic between EC2 and an on-premises server. Which load balancer is appropriate if needed?

A. Application Load Balancer

B. Network Load Balancer

C. Classic Load Balancer

D. Gateway Load Balancer

Question 44 VPC

Which network construct allows multiple VPCs and on-prem networks to interconnect through a central hub?

A. VPC Peering

B. Transit Gateway

C. Direct Connect Gateway

D. PrivateLink

Question 45 VPC

Which logs capture metadata (source/dest IP, port, action) for traffic in a VPC?

A. CloudTrail

B. VPC Flow Logs

C. CloudWatch Logs

D. GuardDuty findings

Question 46 VPC

Which inbound rule on a security group is required for an EC2 web server to receive HTTPS traffic from the internet?

A. Allow TCP 80 from 0.0.0.0/0

B. Allow TCP 443 from 0.0.0.0/0

C. Allow UDP 443 from the VPC CIDR only

D. Allow All Traffic from 0.0.0.0/0

Question 47 VPC

A subnet has 65,536 IP addresses (CIDR /16). Why is the usable count actually less?

A. AWS reserves 5 addresses per subnet (network, VPC router, DNS, future use, broadcast).

B. Half the addresses are reserved for IPv6.

C. The first 1,000 are reserved for ELBs.

D. Reserved Instances consume 100 addresses.

Question 48 VPC

Which feature mirrors network packets from an ENI to a tool for deep packet inspection?

A. VPC Flow Logs

B. VPC Traffic Mirroring

C. AWS Network Firewall

D. Route 53 Resolver query logging

Question 49 DB

Which RDS feature provides automatic failover to a standby in a different AZ with no application code changes?

A. Read Replica

B. Multi-AZ deployment

C. Cross-Region Replication

D. RDS Proxy

Question 50 DB

An application has bursty short-lived connections that exhaust RDS MySQL connection limits. Which AWS service mitigates this?

A. DAX

B. RDS Proxy

C. ElastiCache for Memcached

D. Aurora Serverless v2

Question 51 DB

A NoSQL workload requires single-digit-millisecond reads at any scale, with multi-region active-active writes. Which service?

A. DynamoDB Global Tables

B. Aurora Global Database

C. RDS Multi-AZ

D. Neptune

Question 52 DB

Which DynamoDB feature accelerates read performance to microsecond latency for frequently accessed items?

A. DynamoDB Streams

B. DAX

C. Global Secondary Index

D. On-Demand backups

Question 53 DB

Which is NOT a valid DynamoDB capacity mode?

A. Provisioned

B. On-Demand

C. Reserved

D. Burst (not a mode, but a behavior)

Question 54 DB

Aurora provides storage that scales from 10 GiB to 128 TiB automatically. How is durability achieved?

A. Single AZ with snapshot every 5 minutes.

B. 6 copies of data across 3 AZs; 4-of-6 write quorum, 3-of-6 read quorum.

C. RAID-5 across instance store.

D. Continuous replication to S3 only.

Question 55 DB

An OLAP workload runs 10-second queries over 500 GB. Which service is BEST?

A. RDS PostgreSQL

B. Aurora

C. Redshift

D. DynamoDB

Question 56 DB

ElastiCache for Redis vs Memcached — which supports persistence and pub/sub?

A. Memcached

B. Redis

C. Both

D. Neither

Question 57 DB

Which RDS engine supports cross-region read replicas?

A. MySQL only

B. PostgreSQL only

C. MySQL, PostgreSQL, MariaDB, Oracle

D. All RDS engines including SQL Server

Question 58 DB

Which is the BEST way to migrate a 5 TB Oracle database to Aurora PostgreSQL?

A. Manual export/import via pg_dump

B. AWS DMS for data migration + Schema Conversion Tool (SCT) for schema/code

C. Snapshot Oracle and restore to Aurora directly

D. Use AWS Backup

Question 59 DB

Which DynamoDB index allows a different partition key from the base table?

A. Local Secondary Index

B. Global Secondary Index

C. Both

D. Neither

Question 60 DB

Aurora Serverless v2 differs from v1 in which key way?

A. v2 supports MySQL only

B. v2 scales in fine-grained increments and works with Aurora cluster features like Global DB, Multi-AZ replicas

C. v2 is cheaper than v1 at all times

D. v2 requires a single master always-on

Question 61 SERVERLESS

A Lambda function reads from an SQS queue. The function fails intermittently. What ensures messages are not lost?

A. SQS automatically deletes messages on receive.

B. Lambda only deletes the message after successful invocation; failures return the message to the queue after visibility timeout.

C. Set retention to 0 seconds.

D. Convert to SNS.

Question 62 SERVERLESS

Which AWS service is BEST for orchestrating a multi-step workflow with retries, error handling, and parallel branches?

A. AWS Lambda chained with destinations

B. Amazon SQS

C. AWS Step Functions

D. Amazon EventBridge

Question 63 SERVERLESS

What is the maximum execution duration for a single Lambda invocation?

A. 1 minute

B. 5 minutes

C. 15 minutes

D. 1 hour

Question 64 SERVERLESS

An application produces events that must fan out to multiple consumers (Lambda, SQS, HTTPS endpoints). Which service is appropriate?

A. SQS

B. SNS

C. Kinesis Data Streams

D. EventBridge

Question 65 SERVERLESS

An e-commerce site needs to process orders in strict FIFO order per customer, with no duplicates. Which queue?

A. SQS Standard

B. SQS FIFO with content-based dedup and MessageGroupId per customer

C. SNS Standard topic

D. Kinesis Data Streams

Question 66 SERVERLESS

Lambda function memory is set to 1024 MB. What also scales with memory?

A. Network bandwidth only

B. CPU, network, and disk I/O proportionally

C. Just storage

D. Nothing — memory is independent

Question 67 SERVERLESS

Which feature minimizes Lambda cold starts for latency-sensitive APIs?

A. Reserved Concurrency

B. Provisioned Concurrency

C. Increase memory to 10 GB

D. Use a custom runtime

Question 68 SERVERLESS

API Gateway routes requests to a Lambda. The API must enforce 1000 RPS per API key. Which feature handles this?

A. Usage plans with throttling

B. Lambda Reserved Concurrency

C. WAF rate-based rules

D. CloudFront caching

Question 69 SERVERLESS

What does CloudFront's 'Origin Access Control' (OAC) achieve?

A. Caches dynamic content

B. Allows CloudFront to access a private S3 origin while blocking direct S3 URLs

C. Replicates origin to edge

D. Encrypts traffic between viewer and edge

Question 70 SERVERLESS

A workload processes 1 million events/sec with strict ordering by user ID. Which service?

A. SNS

B. Kinesis Data Streams with shard count tuned and user ID as partition key

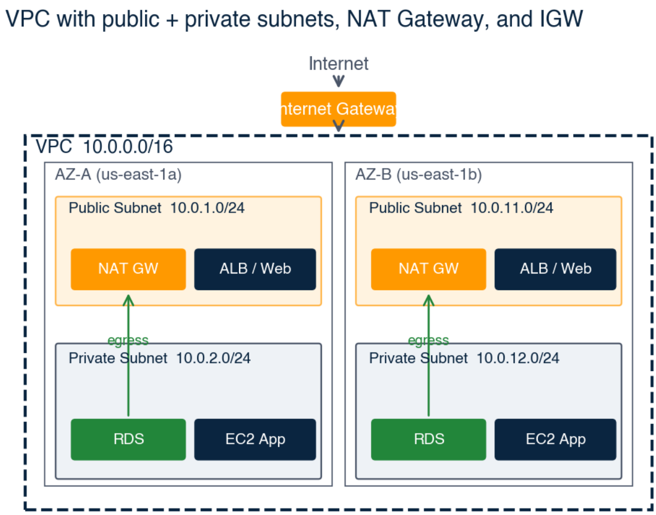
C. SQS FIFO

D. Lambda function URLs

Visual Reference

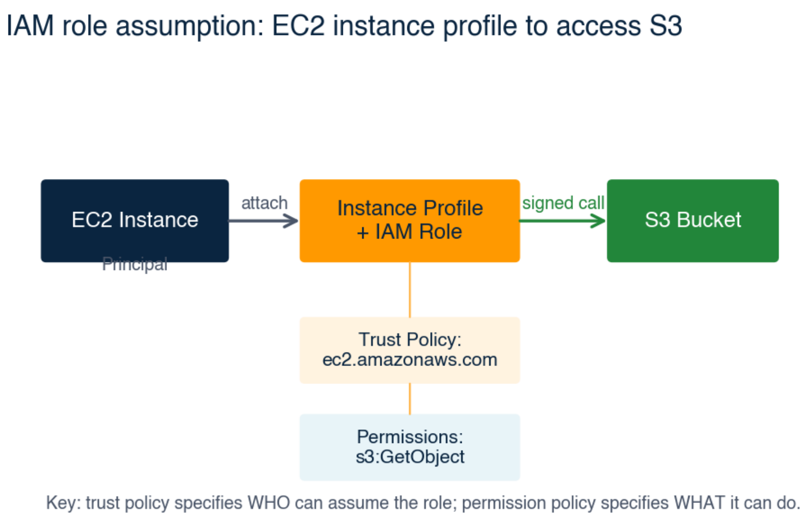
Twelve diagrams covering the patterns the SAA-C03 tests most often. Refer back to these while working questions and answers. The cross-reference under each diagram lists the questions in this volume that depend on the pattern.

VPC with public + private subnets, NAT, and IGW



Relevant questions: Q37 NAT Gateway, Q41 NAT per AZ, Q47 reserved IPs

IAM role assumption: EC2 instance profile



Relevant questions: Q1, Q3 PassRole, Q6 task role vs execution role

S3 storage classes: cost vs retrieval

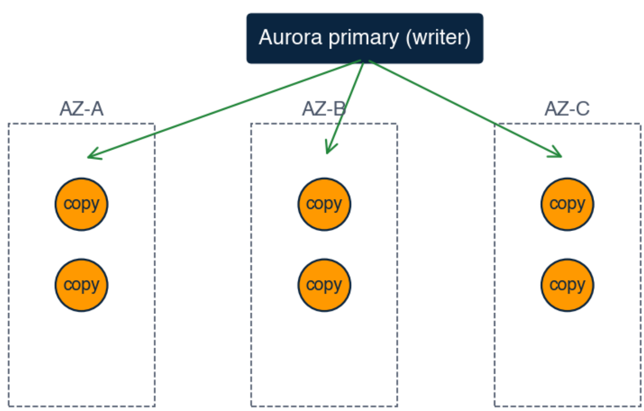
S3 storage classes — cost vs retrieval trade-off

Tier	Min storage	Retrieval	Use case
Standard	none	ms	Hot data, frequent access
Standard-IA	30 days	ms	Infrequent access >30d
One Zone-IA	30 days	ms	Recreatable + cost priority
Glacier Instant	90 days	ms	Archive w/ quick read
Glacier Flexible	90 days	min-hr	Backups, DR copies
Glacier Deep	180 days	12 hr	Compliance archives
Intelligent-Tiering	none	ms	Auto-tiers by access pattern

Relevant questions: Q25 lifecycle, Q30 minimum-storage rules, Q33 KMS

Aurora: 6 copies across 3 AZs

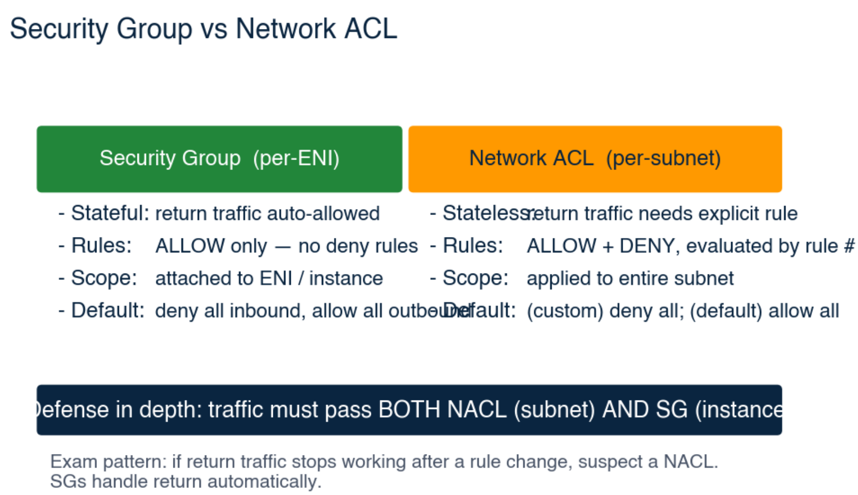
Aurora storage: 6 copies across 3 AZs



Writes replicate to 6 copies across 3 AZs. Quorum: 4-of-6 writes, 3-of-6 reads.
Tolerates loss of 1 AZ + 1 additional copy without data loss; 1 AZ loss without read disruption.

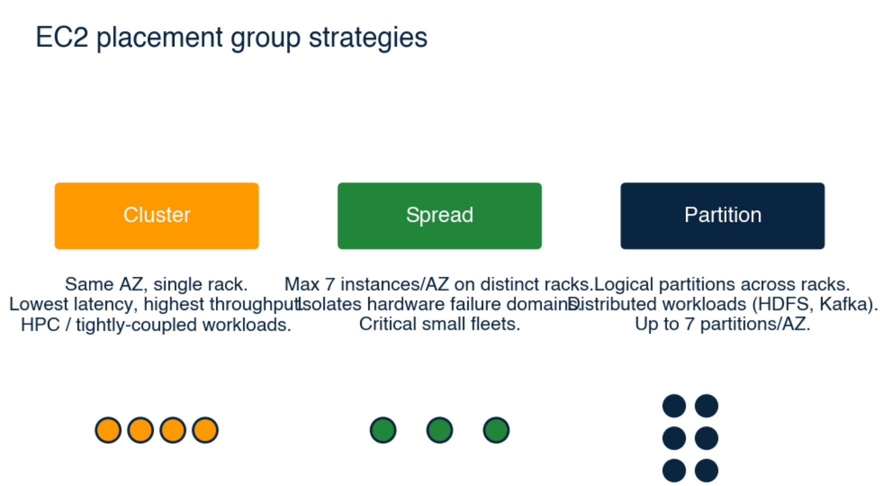
Relevant questions: Q54 Aurora durability, Q60 Serverless v2

Security Group vs Network ACL



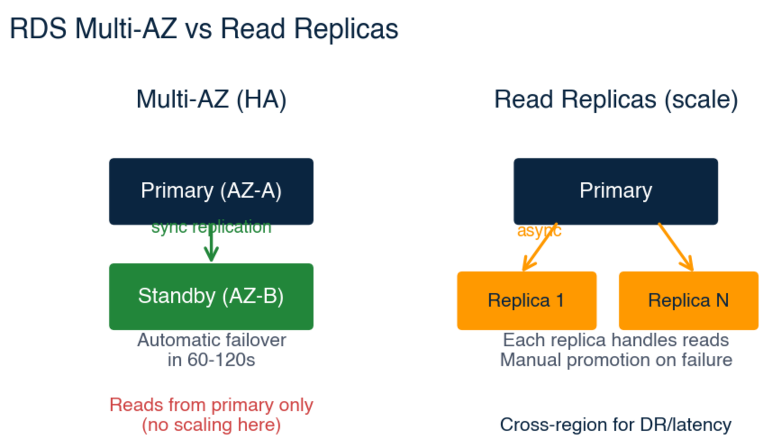
Relevant questions: Q38 SG vs NACL, Q46 inbound HTTPS rule

EC2 placement-group strategies



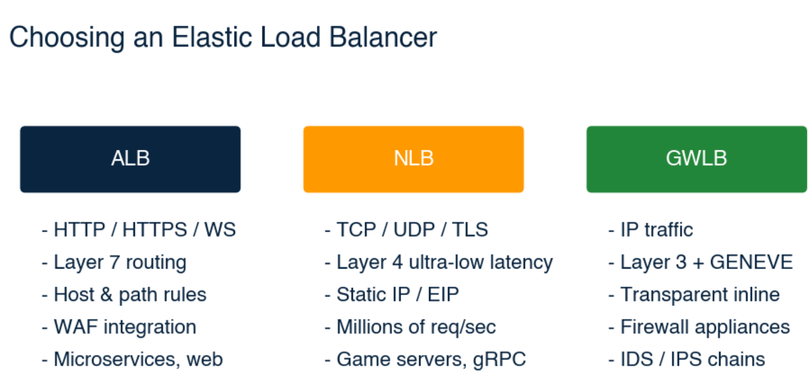
Relevant questions: Q14 cluster placement for HPC

Multi-AZ vs Read Replica



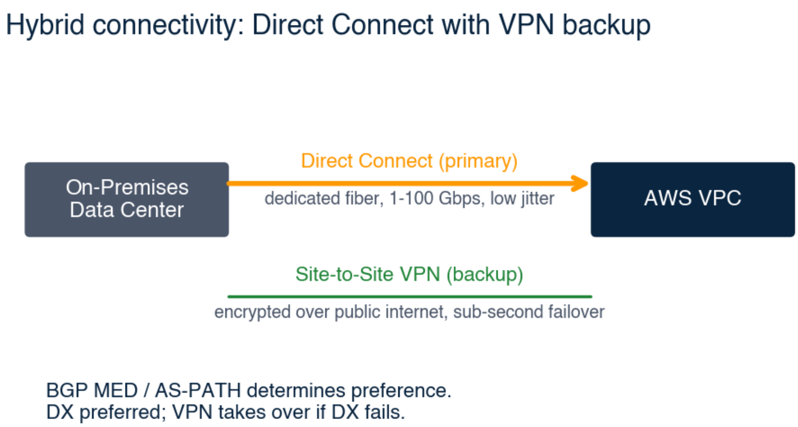
Relevant questions: Q49 Multi-AZ failover, Q57 cross-region replicas

Choosing ALB / NLB / GWLB



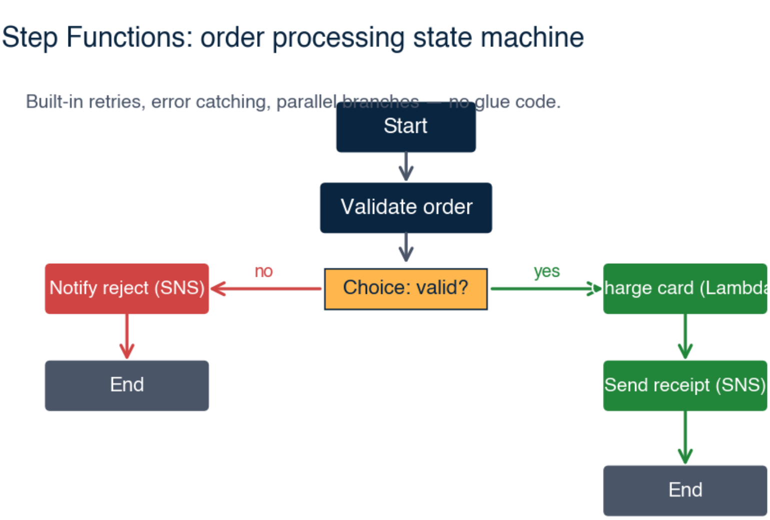
Relevant questions: Q43 UDP traffic, ALB vs NLB selection

Direct Connect with VPN backup



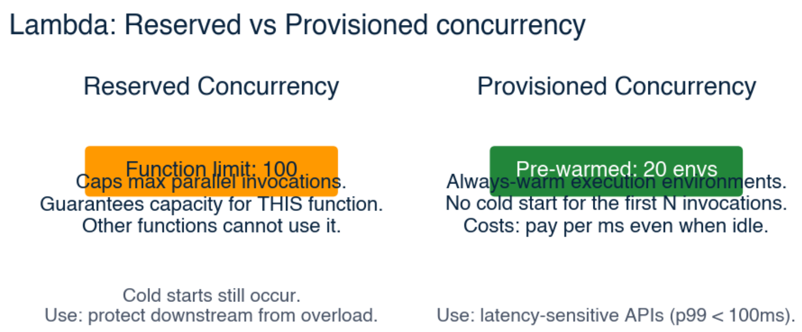
Relevant questions: Q42 hybrid connectivity at 10 Gbps

Step Functions state machine



Relevant questions: Q62 orchestration vs Lambda chaining

Lambda: Reserved vs Provisioned



Relevant questions: Q67 cold-start mitigation

KMS key types and when to choose

KMS key types — when to choose which

Type	Control / cost	Use case
AWS-owned	AWS-managed	S3 default SSE-S3 (transparent)
AWS-managed	Per-service, you can audit	RDS default, ELB cert, etc.
Customer-managed	You control rotation + policy	Cross-account, audit, BYOK
CloudHSM-backed	Hardware in single-tenant HSM	FIPS 140-2 Level 3 compliance

Relevant questions: Q33 SSE-KMS, Q10 cross-key permissions

Answers and Explanations

Read the explanation for every question — including the ones you got right. Many SAA-C03 questions are decided by understanding why the OTHER options are wrong. We mark wrong choices implicitly through explanation context.

Question 1 **Answer: B** *IAM*

EC2 instance profiles deliver short-lived credentials via the metadata service (IMDSv2). No long-lived keys, no rotation overhead, no extra services. Option A violates the policy. C is more work than B for no benefit. D is for human users, not EC2 workloads.

Question 2 **Answer: C** *IAM*

Cross-account IAM roles with sts:AssumeRole avoid sharing any long-lived credential and enforce short session durations. A leaves a window for credential leakage. B is never acceptable. D exposes keys in email.

Question 3 **Answer: B** *IAM*

When attaching an instance profile at launch, the launching principal must have iam:PassRole for that role. This is one of the most-tested gotchas on SAA-C03.

Question 4 **Answer: C** *IAM*

aws:SourceVpce restricts by VPC endpoint ID. aws:SourceVpc restricts by VPC (broader). aws:SourceIp would not work because traffic via a Gateway endpoint uses private IPs that change.

Question 5 **Answer: B** *IAM*

IAM Identity Center is the recommended way to centrally manage workforce access across multiple AWS accounts. Cognito is for application end-users. RAM shares resources, not identities.

Question 6 **Answer: B** *IAM*

Task role: used by application code inside the container. Task execution role: used by the ECS agent to pull images and write logs. Different concerns, often confused.

Question 7 **Answer: B** *IAM*

aws:SecureTransport is true only when TLS is used. This is the canonical pattern for enforcing HTTPS-only on S3 and AWS Security Hub flags buckets without it.

Question 8 **Answer: C** *IAM*

Valid components: Effect, Action, Resource, Principal (resource-based only), Condition, Sid, Version. TargetGroup is an ELB construct, not IAM.

Question 9 **Answer: B** *IAM*

Explicit Deny in ANY applicable policy overrides any Allow. Memorize this: Deny > Allow > implicit Deny.

Question 10 **Answer: B** *IAM*

Customer-managed KMS keys require explicit kms:Decrypt on the key's policy AND the principal's identity policy. AWS-managed keys don't require this dual permission.

Question 11 **Answer: B** *IAM*

Access Advisor shows services the role/user accessed and when — used for tightening least-privilege. Credential Report is about credential lifecycle. Policy Simulator tests theoretical evaluation.

Question 12 **Answer: B** *IAM*

Attribute-Based Access Control (ABAC) via aws:ResourceTag conditions is the standard pattern. SCPs apply to whole accounts, not subsets of resources.

Question 13 **Answer: C** *EC2*

Match commit-style pricing to predictable load, On-Demand/Spot to variable load. Spot alone risks interruption; full RIs waste capacity off-peak.

Question 14 **Answer: A** *EC2*

Cluster: same AZ, single rack — lowest latency. Spread: max isolation, max 7 per AZ. Partition: large distributed workloads like HDFS.

Question 15 **Answer: B** *EC2*

Capacity issues are AZ-specific. Solutions: try a different AZ, switch to a similar instance type, or use Capacity Reservations to guarantee.

Question 16 **Answer: C** *EC2*

io2 Block Express scales to 256,000 IOPS with sub-ms latency — ideal for I/O-intensive databases. gp3 caps at 16,000 IOPS. st1 is throughput-optimized HDD.

Question 17 **Answer: C** *EC2*

Instance store = ephemeral, attached physically to the host. Stop/terminate/hardware failure = data gone. Use EBS for persistence.

Question 18 **Answer: C** *EC2*

Predictive scaling uses ML on historical traffic to provision capacity BEFORE demand arrives. Target tracking and step scaling are reactive.

Question 19 **Answer: B** *EC2*

Dedicated Host: full physical server, you control placement, useful for BYOL licensing. Dedicated Instance: isolation but AWS controls placement. Bare Metal: instance type without hypervisor.

Question 20 **Answer: B** *EC2*

EFS: multi-AZ POSIX file system, scales to thousands of concurrent clients. EBS Multi-Attach is limited to 16 io1/io2 instances in ONE AZ.

Question 21 **Answer: B** *EC2*

Spot Advisor's interruption rate is over the trailing month. >20% means fragile capacity — use diversified pools or switch instance types.

Question 22 **Answer: B** *EC2*

Golden images = build once, deploy many. EC2 Image Builder automates the pipeline. Immutable infra eliminates configuration drift.

Question 23 **Answer: B** *EC2*

ASG rebalances across healthy AZs to maintain desired count — a core resilience feature. This is why you should never run a single-AZ ASG for prod.

Question 24 Answer: B *EC2*

Instance refresh on an ASG with a new launch template version rolls capacity in batches with health checks — zero-downtime updates. The immutable pattern.

Question 25 Answer: B *S3*

Standard for hot tier, Standard-IA for warm (30 days minimum), Flexible Retrieval (formerly Glacier) supports minutes-to-hours retrieval. Deep Archive requires 12 hours and is wrong here.

Question 26 Answer: C *S3*

Object Lock provides WORM (Write Once Read Many). Compliance mode prevents even root from deleting. Versioning preserves old versions but allows deletion.

Question 27 Answer: B *S3*

Multipart upload: parallel parts, retry only failed parts, recommended for >100 MB and required for >5 GB. S3 Transfer Acceleration also helps over long distances.

Question 28 Answer: C *S3*

All S3 tiers offer 11 nines durability. Availability differs: Standard 99.99%, Standard-IA 99.9%, One Zone-IA 99.5%.

Question 29 Answer: B *S3*

Defense in depth: prevent public exposure, encrypt with customer-managed key, force TLS, audit access. Each layer fails closed if another is misconfigured.

Question 30 Answer: D *S3*

Standard-IA and One Zone-IA both have 30-day minimum. Glacier Flexible Retrieval = 90 days. Glacier Deep Archive = 180 days. Standard = no minimum.

Question 31 **Answer: B** *S3*

Transfer Acceleration uses CloudFront edges as ingress points — beneficial for clients geographically distant from the bucket. Same-region traffic gets no benefit.

Question 32 **Answer: A** *S3*

Versioning must be enabled on both. CRR can target buckets in different accounts. Bucket names must be globally unique anyway.

Question 33 **Answer: B** *S3*

SSE-KMS calls KMS for every GetObject/PutObject. S3 Bucket Keys mitigate this by caching data keys at the bucket level — must be explicitly enabled.

Question 34 **Answer: B** *S3*

S3 Select: SQL on a single object (CSV/JSON/Parquet). Athena: SQL across many objects with partitioning. Both serverless, both pay-per-query.

Question 35 **Answer: B** *S3*

S3 supports 3,500 PUT/COPY/POST/DELETE and 5,500 GET/HEAD per prefix per second. 503 means you exceeded; backoff and prefix-spread fix it.

Question 36 **Answer: B** *S3*

MRAP provides a global endpoint that uses AWS Global Accelerator under the hood. CRR replicates data but does not route clients.

Question 37 **Answer: B** *VPC*

NAT Gateway: outbound-only internet for private subnets, fully managed, scales to 100 Gbps. NAT Instance is the older self-managed alternative.

Question 38 **Answer: B** *VPC*

SG: stateful (return traffic auto-allowed), allow rules only, applied per-ENI. NACL: stateless, allow+deny rules, applied per-subnet. Classic exam differentiator.

Question 39 Answer: C *VPC*

Neither VPC Peering nor Transit Gateway supports overlapping CIDRs. PrivateLink exposes a specific service via NLB without routing the full VPC.

Question 40 Answer: B *VPC*

Gateway endpoints: S3 and DynamoDB only, free, route-table based. Interface endpoints: most other AWS services, charged per hour + per GB.

Question 41 Answer: A *VPC*

Best practice: one NAT Gateway PER AZ, with the local AZ's route table pointing to it. Saves on inter-AZ data costs and isolates failure domains.

Question 42 Answer: B *VPC*

Direct Connect: dedicated fiber, 1/10/100 Gbps, predictable latency. VPN: encrypted over public internet, 1.25 Gbps per tunnel ceiling. Direct Connect can be backed up with VPN.

Question 43 Answer: B *VPC*

NLB: TCP, UDP, TLS, static IPs, ultra-low latency. ALB: HTTP/HTTPS only. GWLB: inline appliances. Classic: legacy.

Question 44 Answer: B *VPC*

Transit Gateway: hub-and-spoke for 5,000+ VPCs, VPN, Direct Connect, all in one. Replaces complex mesh of peerings.

Question 45 Answer: B *VPC*

VPC Flow Logs: per-ENI or per-subnet or per-VPC, sent to CloudWatch or S3. Does NOT capture packet payloads — use Traffic Mirroring for that.

Question 46 Answer: B *VPC*

HTTPS = TCP 443. From the internet = 0.0.0.0/0. Avoid 'all traffic'. Port 80 is HTTP (would need separate rule).

Question 47 **Answer: A** *VPC*

AWS reserves the .0, .1, .2, .3, and .255 in each subnet. Memorize: 5 reserved per subnet.

Question 48 **Answer: B** *VPC*

Traffic Mirroring: full packet capture from a source ENI to a target (NLB, ENI, or GLB). Used for IDS, NDR, troubleshooting at L4-L7.

Question 49 **Answer: B** *DB*

Multi-AZ: synchronous standby in another AZ, automatic DNS failover in 60-120s. Read replicas are async and used for read scaling, not HA.

Question 50 **Answer: B** *DB*

RDS Proxy: connection pooling and multiplexing in front of RDS/Aurora. Reduces connection churn and supports IAM auth + failover speed.

Question 51 **Answer: A** *DB*

DynamoDB Global Tables: multi-region, multi-active, last-writer-wins. Aurora Global is one writer per region + cross-region read replicas.

Question 52 **Answer: B** *DB*

DAX: write-through caching layer purpose-built for DynamoDB. Lives in your VPC. Eventual consistency for reads.

Question 53 **Answer: C** *DB*

DynamoDB has Provisioned and On-Demand. Reserved capacity is a billing optimization within Provisioned, not a separate mode.

Question 54 **Answer: B** *DB*

Aurora's distributed storage layer: 6 copies (2 per AZ × 3 AZs), survives 2 copy loss without data loss and 3 copy loss without read disruption.

Question 55 **Answer: C** *DB*

Redshift: columnar, MPP, optimized for analytical queries on TBs-PBs. RDS/Aurora are OLTP (row-based, transactional).

Question 56 **Answer: B** *DB*

Redis: persistence, replication, pub/sub, sorted sets, geospatial. Memcached: multi-threaded, multi-AZ NOT supported, no persistence — pure cache.

Question 57 **Answer: C** *DB*

Cross-region read replicas: MySQL, MariaDB, PostgreSQL, Oracle (BYOL). SQL Server requires Multi-AZ DR or native replication.

Question 58 **Answer: B** *DB*

DMS: data migration with minimal downtime via CDC. SCT: converts schema and stored procedures from Oracle to PostgreSQL dialect.

Question 59 **Answer: B** *DB*

GSI: different PK and sort key, eventually consistent, can be created any time. LSI: same PK, different sort key, must be created at table creation.

Question 60 **Answer: B** *DB*

v2: fast (sub-second) granular scaling in ACUs, full compatibility with Aurora cluster features. v1 had cold starts and limited feature support.

Question 61 **Answer: B** *Serverless*

Lambda+SQS: Lambda removes the message only after returning success. On failure, the message reappears after visibility timeout and retries. Use DLQ after maxReceiveCount.

Question 62 **Answer: C** *Serverless*

Step Functions: visual state machine, built-in retries, parallel, choice, map. Far cleaner than Lambda-chains for non-trivial flows.

Question 63 **Answer: C** *Serverless*

15 minutes (900 seconds). For longer workloads, use Step Functions, ECS Fargate, or EC2.

Question 64 **Answer: B** *Serverless*

SNS: pub/sub fanout, push-based, multiple protocol subscribers. SQS is point-to-point. EventBridge is similar but adds schema registry and content-based routing — pick when business event routing is the use case.

Question 65 **Answer: B** *Serverless*

SQS FIFO: strict ordering per MessageGroupId, exactly-once with dedup window. Standard SQS is at-least-once with best-effort ordering.

Question 66 **Answer: B** *Serverless*

Lambda allocates CPU (and proportional network/I/O) linearly with memory. More memory = more CPU = faster execution = often lower cost despite higher per-ms charge.

Question 67 **Answer: B** *Serverless*

Provisioned Concurrency: pre-warmed execution environments, no cold start. Reserved limits maximum parallel invocations but doesn't warm them.

Question 68 **Answer: A** *Serverless*

API Gateway Usage Plans: per-key rate and burst limits. WAF rate rules are coarser (per IP). Reserved Concurrency limits total parallel Lambdas, not per-key RPS.

Question 69 **Answer: B** *Serverless*

OAC (replaces OAI) lets the S3 bucket policy allow only CloudFront. Direct s3:// URLs return 403. Standard pattern for serving static sites privately.

Question 70 **Answer: B** *Serverless*

Kinesis Data Streams: high-throughput, ordered per partition, hot shard tuning via partition key. SQS FIFO caps at ~3000 msg/s per group. SNS is fanout, not ordered.

Volume 1 Cheat Sheet

IAM EVALUATION

- Explicit Deny > Explicit Allow > Implicit Deny.
- iam:PassRole needed to attach instance profile / Lambda execution role.
- Use IAM Identity Center for multi-account workforce SSO.

EC2 STORAGE

- gp3: general 16K IOPS, 1000 MiBps. io2 Block Express: 256K IOPS, sub-ms.
- Instance store = ephemeral. EBS = persistent.
- EFS: multi-AZ POSIX. EBS Multi-Attach: 16 instances same AZ.

S3 TIERS (30-day minimum applies)

- Standard: hot, no min. Standard-IA: warm, 30d. One Zone-IA: warm + single AZ.
- Glacier Flexible: minutes-to-hours retrieval, 90d min.
- Glacier Deep Archive: 12h retrieval, 180d min, cheapest.

VPC RESERVED IPS

- AWS reserves 5 IPs per subnet: .0, .1, .2, .3, .last (broadcast).
- Security Group = stateful, allow-only, per-ENI.
- NACL = stateless, allow+deny, per-subnet.

DATABASE QUICK PICKS

- OLTP relational: RDS / Aurora. OLAP: Redshift.
- Sub-ms key-value: DynamoDB. Caching: ElastiCache (Redis = features).
- Aurora durability: 6 copies across 3 AZs, 4-of-6 quorum.

SERVERLESS LIMITS

- Lambda max duration: 15 min. Memory: up to 10 GB.
- SQS FIFO: ordered per MessageGroupId, exactly-once.
- API Gateway throttling per-key via Usage Plans.

You finished Volume 1.

If you scored 80% or higher on Volume 1 without reviewing answers, you are tracking toward exam readiness.

If you scored below 70%, restart this volume. Read every explanation, write down the AWS service and the pattern in your own words, then re-test in 48 hours.

Next steps:
• Volume 2: Questions 71-140 — Security, KMS, IAM advanced patterns, WAF, Shield, GuardDuty.
• Volume 3: Questions 141-210 — Resilience, DR strategies, multi-region design.
• Volume 4: Questions 211-280 — High performance, caching, content delivery.
• Volume 5: Questions 281-350 — Cost optimization, FinOps, full-length timed mock exam.

Found an error or have a question? We respond to every reader email.

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